

Literature status: unique asymptotic models need not yield Asymptotic ℓ_p subspaces

Run `fa_banach_001`; source arXiv:math/0110146

Source question

Halbeisen and Odell's paper *On asymptotic models in Banach spaces* asks the following question in its open-problem section.

tion 3.5 would also hold in this relaxed setting.

5.2. Open Problems.

(5.2.1.) X is **asymptotic** ℓ_p (respectively, **asymptotic** c_0) if there exists K so that for all $(e_i)_{i \in n} \in \{X\}_n$, $(e_i)_{i \in n}$ is K -equivalent to the unit vector basis of ℓ_p^n (respectively, ℓ_∞^n) (see [MMT95]). Assume that there exists K and $1 \leq p \leq \infty$ so that if $(e_i)_{i \in \omega}$ is an asymptotic model of X , then $(e_i)_{i \in \omega}$ is K -equivalent to the unit vector basis of ℓ_p (c_0 , if $p = \infty$). Does X contain an asymptotic ℓ_p (or c_0) subspace? The analogous problem for spreading models is also open.

Figure 1: arXiv:math/0110146, page 31, item (5.2.1). The source TeX labels the same item as Problem 6.1.

In words, suppose there are K and $1 \leq p \leq \infty$ such that every asymptotic model of X is K -equivalent to the unit vector basis of ℓ_p , with c_0 replacing ℓ_∞ . Must X contain an asymptotic ℓ_p or c_0 subspace?

Supporting literature status

Argyros, Georgiou, Manoussakis, and Motakis explicitly state that their paper *The complete separation of the two finer asymptotic ℓ_p structures for $1 \leq p < \infty$* provides an answer to the corresponding Halbeisen–Odell problem.

ABSTRACT. For $1 \leq p < \infty$, we present a reflexive Banach space $\mathfrak{X}_{\text{awi}}^{(p)}$, with an unconditional basis, that admits ℓ_p as a unique asymptotic model and does not contain any Asymptotic ℓ_p subspaces. D. Freeman, E. Odell, B. Sari and B. Zheng have shown that whenever a Banach space not containing ℓ_1 , in particular a reflexive Banach space, admits c_0 as a unique asymptotic model then it is Asymptotic c_0 . These results provide a complete answer to a problem posed by L. Halbeisen and E. Odell and also complete a line of inquiry of the relation between specific asymptotic structures in Banach spaces, initiated in a previous paper by the first and fourth authors. For the definition of $\mathfrak{X}_{\text{awi}}^{(p)}$ we use saturation with asymptotically weakly incomparable constraints, a new

Figure 2: arXiv:2106.12323, page 1: the abstract states the finite- p construction and says that, together with the c_0 result of Freeman–Odell–Sari–Zheng, it gives a complete answer to the Halbeisen–Odell problem.

The purpose of this article is to provide an answer to the following problem of Halbeisen and Odell from [20] and is, in particular, the last step towards the complete separation of a list of asymptotic structures from [9]. Given a Banach space X , let $\mathcal{F}_0(X)$ denote the family of normalized weakly null sequences in X and $\mathcal{F}_b(X)$ denote the family of normalized block sequences of a fixed basis, if X has one.

Problem 1. Let X be a Banach space that admits a unique asymptotic model with respect to $\mathcal{F}_0(X)$, or with respect to $\mathcal{F}_b(X)$ if X has a basis. Does X contain an Asymptotic ℓ_p , $1 \leq p < \infty$, or an Asymptotic c_0 subspace?

The following definition from [9] provides a more general setting in which we will

Figure 3: arXiv:2106.12323, page 2: the supporting paper restates the Halbeisen–Odell problem for unique asymptotic models.

for $1 \leq p < \infty$. We prove this in the affirmative and, in particular, we show the following.

Theorem 1.4. For $1 \leq p < \infty$, there exists a reflexive Banach space $\mathfrak{X}_{\text{awi}}^{(p)}$ with an unconditional basis that admits ℓ_p as a uniformly unique joint spreading model with respect to $\mathcal{F}_b(\mathfrak{X}_{\text{awi}}^{(p)})$ and contains no Asymptotic ℓ_p subspaces.

Figure 4: arXiv:2106.12323, page 3: Theorem 1.4 gives the finite- p counterexample.

Proof intuition

The source problem asks whether array-level asymptotic uniqueness forces a global asymptotic game structure in some subspace. The later construction separates these two notions. The norm is saturated with asymptotically weakly incomparable constraints so that arrays of normalized blocks still generate only ℓ_p as an asymptotic model, while each infinite-dimensional subspace contains tree segments whose norm is too small to satisfy an Asymptotic ℓ_p lower estimate. Thus the array property survives, but the hereditary global asymptotic property fails.

The literature answer

Theorem 1. *For each $1 \leq p < \infty$, the finite- p part of Halbeisen–Odell’s Problem 6.1 has a negative answer. There is a reflexive Banach space with an unconditional basis that admits ℓ_p as a unique asymptotic model with respect to normalized block sequences, yet contains no Asymptotic ℓ_p subspace.*

Proof. The supporting paper defines the hereditary properties $(a)_p$ and $(b)_p$, where $(a)_p$ is being Asymptotic ℓ_p , and $(b)_p$ is admitting ℓ_p as a uniformly unique joint spreading model, equivalently as a unique asymptotic model, with respect to $\mathcal{F}_b(X)$ [2]. It then says that the only remaining open question in the separation diagram is whether $(b)_p$ completely separates from $(a)_p$ for $1 \leq p < \infty$, and proves this affirmatively.

Concretely, Theorem 1.4 of [2] states that for every $1 \leq p < \infty$ there exists a reflexive Banach space $\mathfrak{X}_{\text{awi}}^{(p)}$ with an unconditional basis that admits ℓ_p as a uniformly unique joint spreading model with respect to $\mathcal{F}_b(\mathfrak{X}_{\text{awi}}^{(p)})$ and contains no Asymptotic ℓ_p subspaces. By the equivalence between

uniformly unique joint spreading models and unique asymptotic models recalled in the same paper, $\mathfrak{X}_{\text{awi}}^{(p)}$ satisfies the asymptotic-model hypothesis of the Halbeisen–Odell question in the finite- p , block-array setting. The conclusion asked for in the problem fails because the space has no Asymptotic ℓ_p subspace at all.

The abstract of [2] also records the complementary c_0 side: Freeman, Odell, Sari, and Zheng proved that whenever a Banach space not containing ℓ_1 , in particular a reflexive Banach space, admits c_0 as a unique asymptotic model, it is Asymptotic c_0 . Thus the finite- p answer is negative, while the c_0 answer is positive under the standard no- ℓ_1 hypothesis. $\square$

Verification notes and limitations

This packet is a literature-status identification, not a new proof. The supporting paper explicitly names Halbeisen and Odell, restates their problem, and says its results provide a complete answer.

The source paper’s item (5.2.1) is phrased for asymptotic models being K -equivalent to ℓ_p or c_0 . The supporting theorem gives the stronger uniqueness/equivalence-to- ℓ_p hypothesis for normalized block arrays and a reflexive counterexample to the requested Asymptotic ℓ_p subspace conclusion. This settles the finite- p direction in the standard block-array interpretation used by the later paper.

The run lightweight indexes were searched for arXiv:0110146 and the key phrases “asymptotic model”, “Halbeisen”, “Odell”, “normalized weakly null”, and “spreading model”. No existing exact packet for this problem was found. A local primary-source search then found arXiv:2106.12323 via the phrases “Halbeisen and Odell”, “complete answer”, and “unique asymptotic model”.

Human review recommendation

Treat the finite- p part of Halbeisen–Odell Problem 6.1 as already answered negatively by [2]. Keep this packet as duplicate/status memory and do not count it as an original proof from this run. Other open problems in the Halbeisen–Odell section remain outside this packet.

References

- [1] L. Halbeisen and E. Odell, *On asymptotic models in Banach spaces*, Israel J. Math. 139 (2004), 253–291; arXiv:math/0110146.
- [2] S. A. Argyros, A. Georgiou, A. Manoussakis, and P. Motakis, *The complete separation of the two finer asymptotic ℓ_p structures for $1 \leq p < \infty$* , arXiv:2106.12323.
- [3] D. Freeman, E. Odell, B. Sari, and B. Zheng, *On spreading sequences and asymptotic structures*, Trans. Amer. Math. Soc. 370 (2018), no. 10, 6933–6953.